Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d-step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \le \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1-\lambda_c)\,\boldsymbol{\delta x[k]} - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k] f_T[k]$$

d-step measurement delay:

$$\boldsymbol{\delta x[k]} = \delta x[k-d] + \sum_{i=1}^{d} \Delta x_d[k-i] - \sum_{i=1}^{d} a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^{d} \delta x_D[k-i]$$

Ideal control effort (with d-step measurement delay):

$$\bar{\bar{f}}_{dx}[k] = a_x^{-1}[k]\Big\{ \underbrace{\Big[\Delta x_d[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \Delta x_d[k-i]\Big]}_{\Delta x_d^d[k]} + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big]$$
$$- \underbrace{\Big[\delta x_D[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \delta x_D[k-i]\Big]}_{\delta x_D^d[k]} \Big\}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\Big\{a_x[k] f_T[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} a_x[k-i] f_T[k-i]\Big\}}_{\delta x_T^d[k]}$$

Implementable control effort

(unchanged in form; $\hat{a}_x$ is now propagated by the estimated slope $\hat{a}'_x$, below)

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k]\big[f_{dx}[k] - \bar{\bar{f}}_{dx}[k]\big] - \delta x_T^d[k]$$

$$a_x[k]\,\bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i]f_{dx}[k-i]\Big] - \delta x_D^d[k]$$

$$a_x[k]\,f_{dx}[k] = \big(1 + e_{a_x}[k]\,\hat{a}_x^{-1}[k]\big)\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\big(\delta x[k-d] + n_x[k]\big) - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i]f_{dx}[k-i]\Big]\Big\}$$

$$\begin{aligned} a_x[k]\,f_{dx}[k] - a_x[k]\,\bar{\bar{f}}_{dx}[k] = {} & e_{a_x}[k]\,f_{dx}[k] + (1-\lambda_c)n_x[k] \\ & + (1-\lambda_c)\textstyle\sum_{i=1}^{d}\big(a_x[k-i] - \hat{a}_x[k-i]\big)f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

(gain slope a'_x promoted to an estimated state — no wall model $c(\bar{h})$)

$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]$; $\quad a_{x_{ram}}[k] \approx a'_x[k]\,x_{ram}[k]$

$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k]$, $\quad \delta a_x[k] = a'_x[k]\,\Delta h_d[k]$, $\quad \delta a_{ram}[k] = a'_x[k]\,\Delta x_{ram}[k]$

$\Delta h_d[k] = h_d[k+1] - h_d[k]$ (desired-trajectory height increment, *known*).

The 4-state takes a'_x from the known wall model. Here a'_x is *unknown*: it is modelled as a slowly-varying state and learned from a_{xm}:

$$\boxed{\begin{cases} a_x[k+1] = a_x[k] + a'_x[k]\,\Delta h_d[k] + \delta a_{ram}[k] \\ a'_x[k+1] = a'_x[k] + w_{a'}[k] \end{cases}}$$

Parameter estimator:

$$\begin{cases} \hat{a}_x[k+1] = \hat{a}_x[k] + \hat{a}'_x[k]\,\Delta h_d[k] \\ \hat{a}'_x[k+1] = \hat{a}'_x[k] \end{cases}$$

Parameter estimation error: $\quad e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$, $\quad e_{a'}[k] = a'_x[k] - \hat{a}'_x[k]$

Estimation error dynamics:

$$\begin{cases} e_{a_x}[k+1] = e_{a_x}[k] + \Delta h_d[k]\,e_{a'}[k] + \delta a_{ram}[k] \\ e_{a'}[k+1] = e_{a'}[k] + w_{a'}[k] \end{cases}$$

(slope $e_{a'}$ taken constant over the short delay window; $h_d[k] - h_d[k-i] = \sum_{j=1}^{i} \Delta h_d[k-j]$)

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - \big(h_d[k] - h_d[k-i]\big)\, e_{a'}[k] - \sum_{j=1}^{i} \delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{f}}_{dx}[k] \approx e_{a_x}[k]\ & \underbrace{\left\{ f_{dx}[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ & - e_{a'}[k]\ \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} \big(h_d[k] - h_d[k-i]\big)\, f_{dx}[k-i] \right\}}_{dF^{h}_{dx}[k]} \\ & - \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} (d+1-i)\, f_{dx}[k-i]\, \delta a_{ram}[k-i] \right\}}_{\delta x^{d}_{\delta af}[k]} \\ & + \delta x^{d}_{D}[k] + (1-\lambda_c) n_x[k] \end{aligned}$$

$$\boxed{\begin{aligned} \delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF^{h}_{dx}[k] \cdot e_{a'}[k] - \delta x^{d}_{D}[k] \\ - \underbrace{\left\{ \delta x^{d}_{T}[k] - \delta x^{d}_{\delta af}[k] + (1-\lambda_c) n_x[k] \right\}}_{\varepsilon[k]} \end{aligned}}$$

State equation

($\delta x^{d}_{D} \equiv 0$: disturbance not modelled)

$$\begin{aligned} \delta x_1[k+1] &= \delta x_2[k] \\ \delta x_2[k+1] &= \delta x_3[k] \\ \delta x_3[k+1] &= \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF^{h}_{dx}[k] \cdot e_{a'}[k] - \varepsilon[k] \\ a_x[k+1] &= a_x[k] + a'_x[k]\, \Delta h_d[k] + \delta a_{ram}[k] \\ a'_x[k+1] &= a'_x[k] + w_{a'}[k] \end{aligned}$$

Output equation

($a_x[k-d] = a_x[k] - \sum_{i=1}^{d} \big(a'_x[k-i] \Delta h_d[k-i] + \delta a_{ram}[k-i]\big)$; slope const $\Rightarrow$ deterministic part $\approx (h_d[k] - h_d[k-d])\, a'_x[k]$)

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} \approx a_x[k] - \underbrace{\big(h_d[k] - h_d[k-d]\big)}_{\Delta H_d[k]\ \text{(known)}}\, a'_x[k] + \underbrace{\left\{ n_a[k] - \textstyle\sum_{i=1}^{d} \delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\Delta H_d[k] \end{bmatrix}, \qquad \Delta H_d[k] = h_d[k] - h_d[k-d], \qquad \mathbf{y}[k] = \mathbf{H}[k]\, \mathbf{x}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta\hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{a'}[k] = a'_x[k] - \hat{a}'_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \hat{a}_x[k] + \Delta H_d[k]\,\hat{a}'_x[k] \\ &= e_{a_x}[k] - \Delta H_d[k]\,e_{a'}[k] + \underbrace{\Big\{n_a[k] - \textstyle\sum_{i=1}^{d} \delta a_{ram}[k-i]\Big\}}_{r_2[k]} \end{aligned}$$

$$\begin{aligned} \delta\hat{x}_1[k+1] &= \delta\hat{x}_2[k] + \ell_{11}[k]\,e_{y_1}[k] + \ell_{12}[k]\,e_{y_2}[k] \\ \delta\hat{x}_2[k+1] &= \delta\hat{x}_3[k] + \ell_{21}[k]\,e_{y_1}[k] + \ell_{22}[k]\,e_{y_2}[k] \\ \delta\hat{x}_3[k+1] &= \lambda_c \cdot \delta\hat{x}_3[k] + \ell_{31}[k]\,e_{y_1}[k] + \ell_{32}[k]\,e_{y_2}[k] \\ \hat{a}_x[k+1] &= \hat{a}_x[k] + \hat{a}'_x[k]\,\Delta h_d[k] + \ell_{41}[k]\,e_{y_1}[k] + \ell_{42}[k]\,e_{y_2}[k] \\ \hat{a}'_x[k+1] &= \hat{a}'_x[k] + \ell_{51}[k]\,e_{y_1}[k] + \ell_{52}[k]\,e_{y_2}[k] \end{aligned}$$

Error dynamics

$$\begin{aligned} e_{\delta x_1}[k+1] &= e_{\delta x_2}[k] - \ell_{11}[k]\,e_{\delta x_1}[k] - \ell_{12}[k]\,e_{a_x}[k] + \ell_{12}[k]\,\Delta H_d[k]\,e_{a'}[k] \\ e_{\delta x_2}[k+1] &= e_{\delta x_3}[k] - \ell_{21}[k]\,e_{\delta x_1}[k] - \ell_{22}[k]\,e_{a_x}[k] + \ell_{22}[k]\,\Delta H_d[k]\,e_{a'}[k] \\ e_{\delta x_3}[k+1] &= \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k]\,e_{a_x}[k] + dF^h_{dx}[k]\,e_{a'}[k] \\ &\quad - \ell_{31}[k]\,e_{\delta x_1}[k] - \ell_{32}[k]\,e_{a_x}[k] + \ell_{32}[k]\,\Delta H_d[k]\,e_{a'}[k] - \underbrace{\varepsilon[k]}_{q_3[k]} \\ e_{a_x}[k+1] &= e_{a_x}[k] + \Delta h_d[k]\,e_{a'}[k] - \ell_{41}[k]\,e_{\delta x_1}[k] - \ell_{42}[k]\,e_{a_x}[k] + \ell_{42}[k]\,\Delta H_d[k]\,e_{a'}[k] + \underbrace{\delta a_{ram}[k]}_{q_4[k]} \\ e_{a'}[k+1] &= e_{a'}[k] - \ell_{51}[k]\,e_{\delta x_1}[k] - \ell_{52}[k]\,e_{a_x}[k] + \ell_{52}[k]\,\Delta H_d[k]\,e_{a'}[k] + \underbrace{w_{a'}[k]}_{q_5[k]} \end{aligned}$$

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF^h_{dx}[k] \\ 0 & 0 & 0 & 1 & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}}$$

$$F_{dx}[k] = f_{dx}[k] + (1-\lambda_c)\sum_{i=1}^{d} f_{dx}[k-i], \qquad dF^h_{dx}[k] = (1-\lambda_c)\sum_{i=1}^{d} \big(h_d[k] - h_d[k-i]\big)\, f_{dx}[k-i]$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}[k]\,\mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k]\,\mathbf{r}[k]$$

Process noise (nonzero on rows 3–5): $\varepsilon[k]$ (lumped thermal / measurement, zero-mean), $\delta a_{ram}[k] = a'_x\,\Delta x_{ram}[k]$, and $w_{a'}[k]$. $q_4 = \delta a_{ram}$ sets the a_x random-walk drive; $q_5 = w_{a'}$ sets the a'_x drive (the estimator bandwidth on the slope). With a'_x now carrying the deterministic gain change, q_4 no longer needs to absorb it.

Remarks — observability of a'_x

The slope a'_x couples to the rest of the state **only through the known height motion**:

$$\mathbf{F}_e(4,5) = \Delta h_d[k] \text{ (prediction path)}, \qquad \mathbf{H}(2,5) = -\Delta H_d[k] \text{ (measurement path)},$$

and both vanish when the desired trajectory holds its height ($\Delta h_d = 0 \Rightarrow \Delta H_d = 0$). Hence:

- a'_x **is structurally unobservable during a height hold**; it is identified only while $\Delta h_d \neq 0$. This is intrinsic, not a filter artifact: a slope can only be read from the gain sampled at two or more heights. (a_x itself, by contrast, is observable from a_{xm} at a single fixed height.)
- Conditioning improves with the height span covered and with slower motion (longer dwell averages down the a_{xm} noise floor), so estimability is strongest under a slow, wide height sweep and weakest under fast oscillation.
- a_{xm} is the **only** feedback channel for a'_x. Its marginal variance $\mathrm{Var}(a_{xm})$ is a function of the instantaneous gain alone and carries **no** slope information; the slope lives in the correlation between $a_{xm}[k]$ and $h_d[k]$. No auxiliary measurement is required or helpful.
- a_{xm} carries a near-wall dynamic bias (the C_{dpmr} closed form degrades when $\hat{a}_x \neq a_x$); since this bias is height-dependent it manifests as an *apparent* slope. Separating the true a'_x from this channel artifact is the central difficulty of the near-wall regime.

Relation to the 4-/5-state variants

variant	gain-rate model	$\mathbf{F}_e(4,5)$	$\mathbf{H}(2,5)$
4-state (a'_x *known*)	$\delta a_x = a'_x \Delta h_d$ fed forward	— (no state)	—
5-state (increment δa_x est.)	$\delta a_x[k{+}1] = \delta a_x[k]$ (free)	1	$-d$
this variant (a'_x *est.*)	$\delta a_x = a'_x \Delta h_d$, a'_x random walk	$\Delta h_d[k]$	$-\Delta H_d[k]$

The 5-state lets the increment δa_x drift with no tie to the trajectory (hence the near-wall lag). This variant reconstructs the increment as $a'_x\,\Delta h_d$ with the *known* regressor Δh_d, factoring the known kinematics out of the unknown wall property — gain-rate tracking becomes parameter identification of a'_x.

Estimability test (planned)

In simulation the true slope $a'_x(h) = -a_x\,K_h/R$ is known, so the estimator is graded directly against it (the truth is used only for scoring; the estimator never sees $c(\bar{h})$). Plan: **(i)** run the slope estimator *open-loop* (estimate $\hat{a}'_x$ but feed the model a'_x, not $\hat{a}'_x$, to the control) to isolate "is the information there" from the closed-loop self-consistency effect; **(ii)** bracket the excitation from passive 1 Hz oscillation to a dedicated slow height sweep, reporting $\hat{a}'_x$ accuracy vs. excitation and vs. height (far-field vs. near-wall); **(iii)** then close the feed-forward ($\delta a_x = \hat{a}'_x \Delta h_d$) and check tracking.